MATH 111 Calculus 1

Inverse trigonometric

and hyperbolic functions

Week 6 · Lecture 14 · §6.6, §6.7

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

STAND UP

"For every real x: $\sin^{-1}(\sin x) = x$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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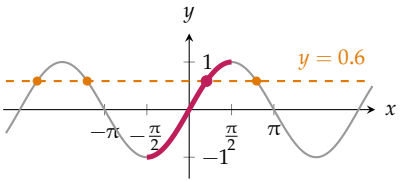
Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. At $x = \pi$: $\sin \pi = 0$, and $\sin^{-1} 0 = 0 \neq \pi$. $\sin^{-1}$ gives only numbers in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so the equation holds exactly when $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (frame 4: undoing).

The inverse sine function

sin is not one-to-one: the line $y = 0.6$ meets its graph again and again.

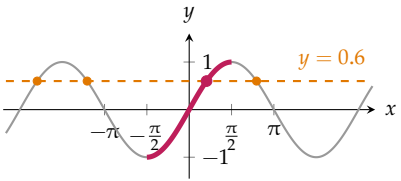


thick: sin on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ — the line meets it once

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Restrict first, then invert. On $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the sine is increasing, and takes every value in $[-1, 1]$ exactly once.



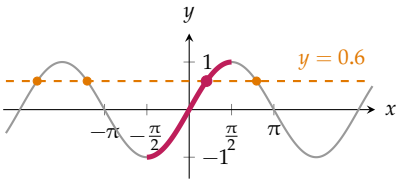
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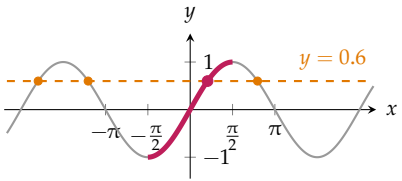
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$\sin^{-1} x$ (also written $\arcsin x$) is the number in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ whose sine is x. It is not $\frac{1}{\sin x}$.

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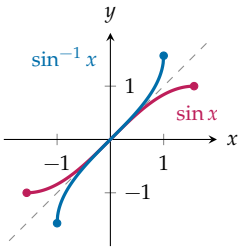
EXAMPLES $\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$, since $\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$ and $\frac{\pi}{3}$ is in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$ — not $\frac{7\pi}{6}$: $\sin\frac{7\pi}{6} = -\frac{1}{2}$ too, but $\frac{7\pi}{6}$ is not in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Undoing, and the graph of $\sin^{-1}$

Undoing $\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$; $\sin(\sin^{-1} x) = x$ for $-1 \le x \le 1$.

Outside $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the first one fails — the warm-up: $\sin^{-1}(\sin \pi) = 0$.



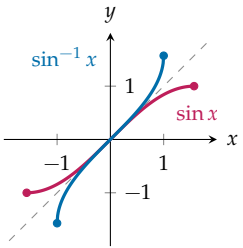
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Restrict first, then invert: the graph of $\sin^{-1}$ is the restricted sine reflected in $y = x$. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; increasing; continuous, as the inverse of a continuous one-to-one function on an interval.



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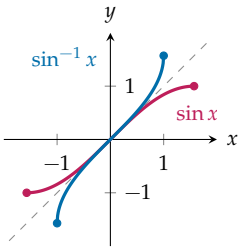
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EXAMPLE $\tan(\sin^{-1} \frac{8}{17})$. Put $\theta = \sin^{-1} \frac{8}{17}$: $\sin\theta = \frac{8}{17}$ and $0 < \theta < \frac{\pi}{2}$.



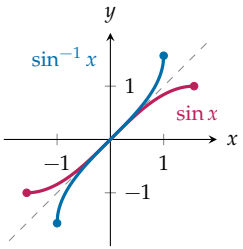
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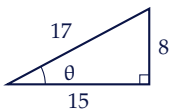
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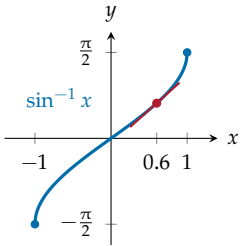
EXAMPLE $\tan(\sin^{-1} \frac{8}{17})$. Put $\theta = \sin^{-1} \frac{8}{17}$: $\sin\theta = \frac{8}{17}$ and $0 < \theta < \frac{\pi}{2}$.

A right triangle: hypotenuse 17, opposite side 8; the third side is $\sqrt{17^2 - 8^2} = 15$. So $\tan(\sin^{-1} \frac{8}{17}) = \frac{8}{15}$.



The derivative of $\sin^{-1}$

Theorem $\dfrac{d}{dx}\sin^{-1}x = \dfrac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$ ($\sin^{-1}$ is differentiable there).

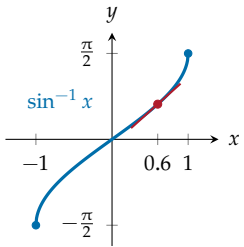


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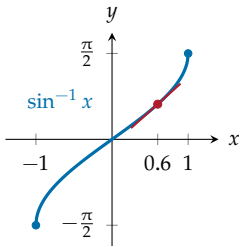
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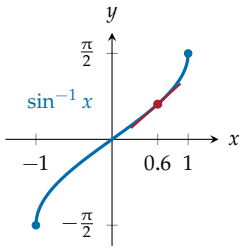
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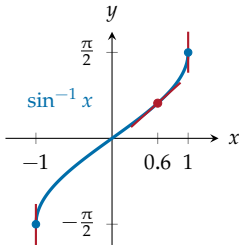
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At $x = \pm 1$: $\cos y = 0$ — no derivative; vertical tangents, the mirror images of the horizontal tangents of sin at $\pm\frac{\pi}{2}$.



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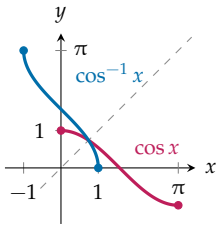
The inverse cosine, briefly

Definition For $-1 \le x \le 1$: $\cos^{-1} x = y \iff \cos y = x$ and $0 \le y \le \pi$.

Undoing $\cos^{-1}(\cos x) = x$ for $0 \le x \le \pi$; $\cos(\cos^{-1} x) = x$ for $-1 \le x \le 1$.

cos is decreasing on $[0, \pi]$, from 1 to -1:
$\cos a - \cos b = 2 \sin \frac{a+b}{2} \sin \frac{b-a}{2} > 0$ for $0 \le a < b \le \pi$.

EXAMPLE $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$: $\cos \frac{2\pi}{3} = -\frac{1}{2}$, and $\frac{2\pi}{3}$ is in $[0, \pi]$.



cos on $[0, \pi]$, reflected in $y = x$

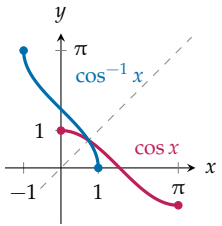
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Theorem $$\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

PROOF As for $\sin^{-1}$, now with $\cos' y = -\sin y \neq 0$ on $(0, \pi)$:
$y = \cos^{-1} x$, $\cos y = x$, $-\sin y \cdot y' = 1$; for $0 < y < \pi$,
$\sin y > 0$, so $\sin y = \sqrt{1-x^2}$ and $y' = -\frac{1}{\sqrt{1-x^2}}$. ■



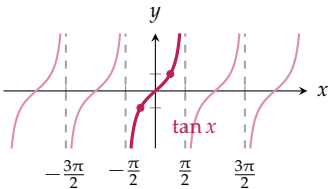
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The inverse tangent function

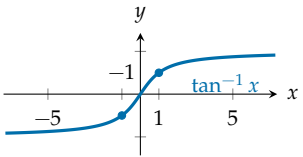
Definition $\tan^{-1} x = y \iff \tan y = x$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$.
Domain $\mathbb{R}$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

tan is increasing on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: $\tan b - \tan a = \frac{\sin(b-a)}{\cos a \cos b} > 0$ for $a < b$; it takes every real value ($\lim_{y \to \pi/2^-} \tan y = \infty$, $\lim_{y \to -\pi/2^+} \tan y = -\infty$).

For example, $\tan^{-1} 1 = \frac{\pi}{4}$, $\tan^{-1}(-1) = -\frac{\pi}{4}$.



$\tan x$, period π; thick: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, one-to-one



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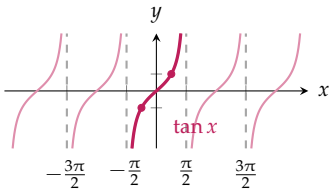
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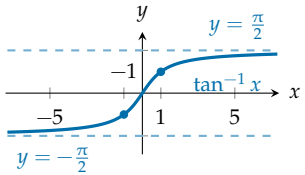
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Limits $\lim_{x \to -\infty} \tan^{-1} x = -\frac{\pi}{2}$, $\lim_{x \to \infty} \tan^{-1} x = \frac{\pi}{2}$.

Given $0 < \varepsilon < \frac{\pi}{2}$: $\frac{\pi}{2} - \varepsilon < \tan^{-1} x < \frac{\pi}{2}$ once $x > \tan\left(\frac{\pi}{2} - \varepsilon\right)$, as $\tan^{-1}$ is increasing; the same at $-\infty$.



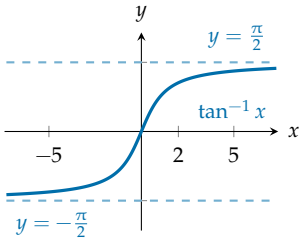
tan x, period π; thick: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, one-to-one



dashed: the horizontal asymptotes

The derivative of $\tan^{-1}$

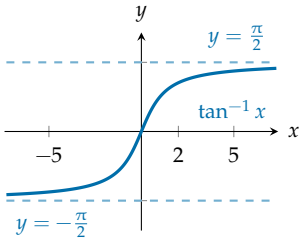
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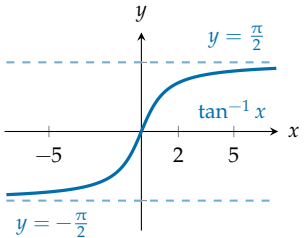


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Let $y = \tan^{-1} x$, so $\tan y = x$. Differentiate implicitly: $\sec^2 y \cdot y' = 1$, so

$$y' = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}.$$



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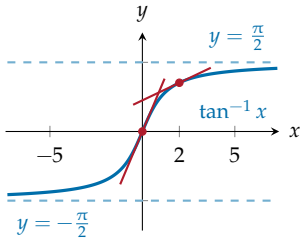
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Let $y = \tan^{-1} x$, so $\tan y = x$. Differentiate implicitly: $\sec^2 y \cdot y' = 1$, so

$$y' = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}.$$

The slope $\frac{1}{1+x^2}$ is largest at $x = 0$, where it is 1, and tends to 0 far out: the graph flattens towards its asymptotes. At $x = 2$ the slope is $\frac{1}{5}$.



red: the tangents at 0 (slope 1) and at 2 (slope $\frac{1}{5}$)

Table of derivatives of inverse trigonometric functions

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}} \qquad \text{for } -1 < x < 1$$

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$$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2} \qquad \text{for every real } x$$

Domains: $\sin^{-1}, \cos^{-1}$: $[-1,1]$; $\tan^{-1}$: $\mathbb{R}$.

At $x = \pm 1$ the graphs of $\sin^{-1}$ and $\cos^{-1}$ have vertical tangents: no derivative there.

The table, and the chain rule

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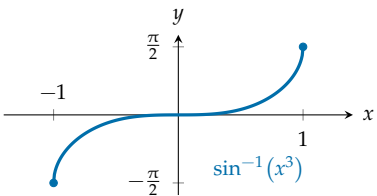
With the chain rule ($u = g(x)$ differentiable):

$$\frac{d}{dx}\sin^{-1}u = \frac{u'}{\sqrt{1-u^2}} \quad (|u| < 1), \qquad \frac{d}{dx}\tan^{-1}u = \frac{u'}{1+u^2}.$$

The chain rule with $\sin^{-1}$: an example

EXAMPLE $y = \sin^{-1}(x^3)$, that is $y = \underbrace{\sin^{-1} u}_{\text{outside function}}$ with $u = \underbrace{x^3}_{\text{inside function}}$; defined for $-1 \le x \le 1$,

as $|x^3| \le 1$ exactly when $|x| \le 1$.

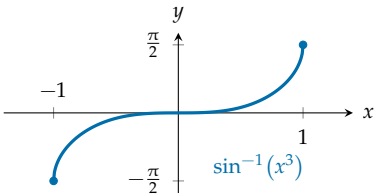


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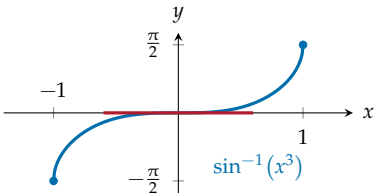
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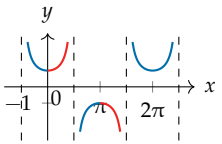
The picture agrees: at $x = 0$ the slope is 0 (the factor $3x^2$), and near $x = \pm 1$ it grows without bound (the root in the denominator tends to 0).



Three more inverse trigonometric functions

Definition

$$y = \csc^{-1} x \ \ (|x| \geq 1) \iff \csc y = x \text{ and } y \in \left(0, \tfrac{\pi}{2}\right] \cup \left(\pi, \tfrac{3\pi}{2}\right]$$
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$y = \sec x$; red: our intervals

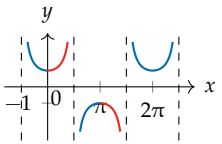
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Our choice, in the box, makes the derivative simpler: no $|x|$. Some authors take $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ for $\sec^{-1}$; then $\sec^{-1}(-2) = \frac{2\pi}{3}$.

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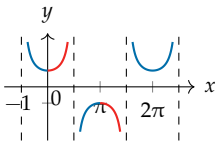
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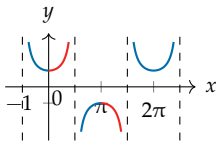
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$\left(\csc^{-1} x\right)'$, $\left(\cot^{-1} x\right)'$ in the same way: homework. $\cot^{-1}(-1) = \frac{3\pi}{4}$.



$y = \sec x$; red: our intervals

Question 38

$$\frac{d}{dx}\tan^{-1}(3x) = \ ?$$

A. $\dfrac{1}{1+9x^2}$

B. $\dfrac{3}{1+9x^2}$

C. $\dfrac{3}{1+x^2}$

D. $3\cos^2(3x)$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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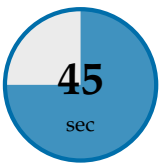
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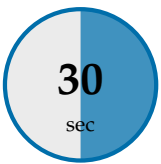
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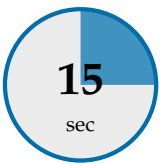
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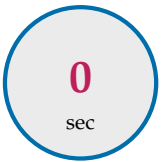
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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Hands down. Thank you.

A B C D

Question 38

EXPLAIN

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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$\frac{d}{dx}\tan^{-1}(3x) = \ ?$

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Answer

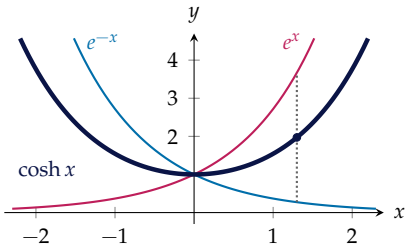
B. Outside $\tan^{-1}$, inside $3x$: $\frac{1}{1+(3x)^2}\cdot 3 = \frac{3}{1+9x^2}$. A forgets the inside's derivative, 3. C evaluates the outside's derivative $\frac{1}{1+u^2}$ at x, not at the inside $3x$. D takes $(\tan^{-1})'(u)$ to be $\frac{1}{\tan'(u)} = \cos^2 u$: the derivative of tan at u, not at $\tan^{-1} u$. At $x = \frac{1}{3}$ the slope is 1.5 (B); C gives 2.7, D gives $3\cos^2 1 \approx 0.88$.

Definition of the hyperbolic functions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x};$$
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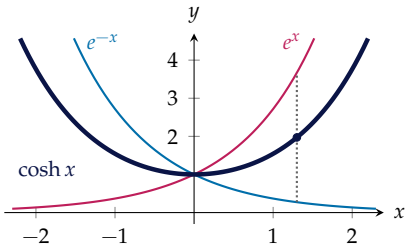


dotted: from $e^{-1.3}$ to $e^{1.3}$; the dot is the midpoint

$\cosh x$ is the average of e^x and e^{-x}: halfway between the two curves. It is even, and its lowest value is $\cosh 0 = 1$.

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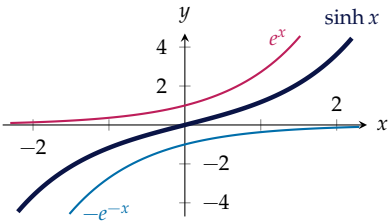
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Range of cosh: $[1, \infty)$.

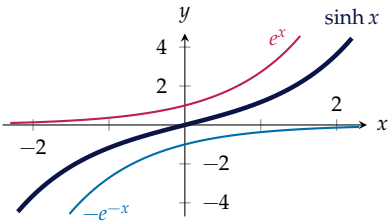
$\cosh x - 1 = \frac{1}{2}\left(e^{x/2} - e^{-x/2}\right)^2 \geq 0$; and every value ≥ 1 is taken ($\cosh 0 = 1$ and $\lim_{x\to\infty} \cosh x = \infty$).

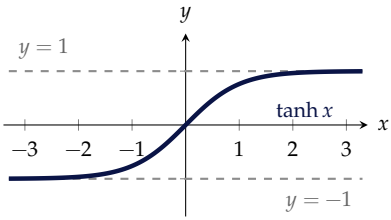
The graphs of sinh and tanh



$\sinh x$: the average of e^x and $-e^{-x}$. Odd, increasing (e^x increases, e^{-x} decreases); range $\mathbb{R}$. $\lim_{x\to\pm\infty} \sinh x = \pm\infty$.

The graphs of sinh and tanh



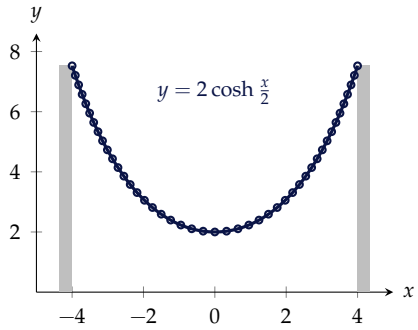


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$\tanh x = \dfrac{1-e^{-2x}}{1+e^{-2x}}$ (divide by e^x):

$\lim_{x\to\infty} \tanh x = \lim_{x\to\infty} \dfrac{1-e^{-2x}}{1+e^{-2x}} = 1$, and $\lim_{x\to-\infty} \tanh x = -1$. Range $(-1,1)$ ($|1-e^{-2x}| < 1+e^{-2x}$); horizontal asymptotes $y = \pm 1$.

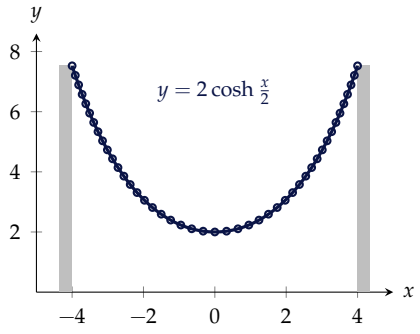
The hanging chain



x and y in metres; grey: the two posts

A heavy uniform flexible chain, hung between two points at the same height, takes the shape $y = c + a\cosh\frac{x}{a}$ (constants $a > 0$ and c): a catenary (Latin *catena*, chain). Taken without proof: it needs the balance of the forces on each piece of the chain.

The hanging chain

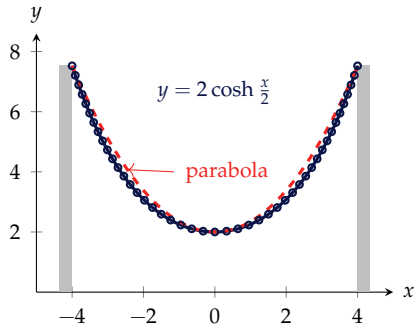


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Our chain: posts 8 m apart; we take $a = 2$ — a chain $4\sinh 2 \approx 14.5$ m long (an arc length: taken without proof) — and $c = 0$: $y = 2\cosh\frac{x}{2}$. The lowest point is 2 m above the ground; the posts are $2\cosh 2 \approx 7.52$ m tall.

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Not a parabola. The red dashed parabola through the same three points, $y = 2 + 0.345\,x^2$, is about 0.29 m higher at $x = \pm 2$: 3.38 against $2\cosh 1 \approx 3.09$.

The identity, and the name

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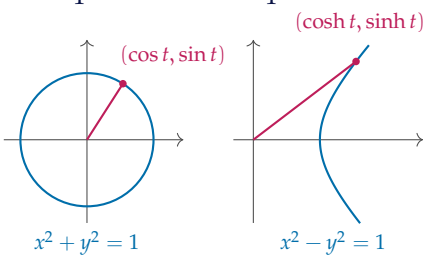
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The name. $(\cos t, \sin t)$ lies on the circle $x^2 + y^2 = 1$: *circular* functions. $(\cosh t, \sinh t)$ lies on the hyperbola $x^2 - y^2 = 1$ — on its right branch, since $\cosh t \geq 1$: *hyperbolic* functions.



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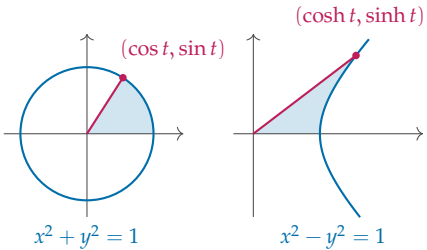
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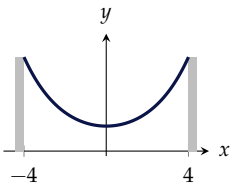
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In both pictures t is twice the area of the shaded sector (for the hyperbola: taken without proof). On the hyperbola, t is not an angle.



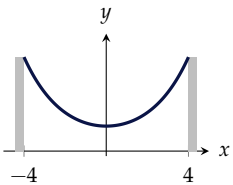
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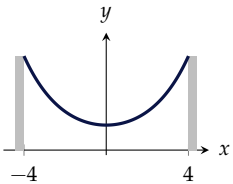


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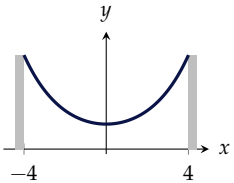
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EXAMPLE The chain of frame 17, $y = 2\cosh\left(\frac{x}{2}\right)$:

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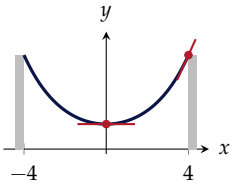
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At the lowest point, $x = 0$: $y' = \sinh 0 = 0$. At the right post, $x = 4$: $y' = \sinh 2 \approx 3.63$ — there the chain rises 3.63 m per metre.



red: tangents at 0 and 4

Question 39

$\dfrac{d}{dx}\tanh x = \ ?$ Here $\tanh x = \dfrac{\sinh x}{\cosh x}$, $\coth x = \dfrac{\cosh x}{\sinh x}$, $\operatorname{sech} x = \dfrac{1}{\cosh x}$.

A. $1 + \tanh^2 x$

B. $\coth x$

C. $\operatorname{sech} x$

D. $1 - \tanh^2 x$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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60 sec

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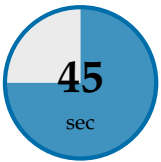
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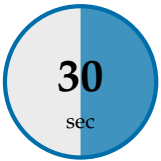
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Hands down. Thank you.

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

D. Quotient rule: $\frac{\cosh x \cdot \cosh x - \sinh x \cdot \sinh x}{\cosh^2 x} = 1 - \frac{\sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x$ $\left(= \operatorname{sech}^2 x,\right.$ by $\left.\cosh^2 x - \sinh^2 x = 1\right)$. A takes $(\cosh x)' = -\sinh x$, the sign of cos: $\frac{\cosh^2 x + \sinh^2 x}{\cosh^2 x}$. B divides the two derivatives: $\frac{\cosh x}{\sinh x}$. C forgets to square the denominator: $\frac{1}{\cosh x}$.

Table of derivatives of hyperbolic functions

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Beside trigonometry: $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x = 1 + \tan^2 x$.

Derivatives: hyperbolic and trigonometric

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The same pattern; the signs differ where the identities differ: $\cos^2 x + \sin^2 x = 1$, but $\cosh^2 x - \sinh^2 x = 1$.

Inverse hyperbolic functions

sinh is increasing (e^x up, e^{-x} down) and onto $\mathbb{R}$: it has an inverse. So do tanh ($\tanh x = 1 - 2/(e^{2x}+1)$) and cosh on $[0,\infty)$ ($\cosh b - \cosh a = 2\sinh\frac{a+b}{2}\sinh\frac{b-a}{2} > 0$, $0 \le a < b$).

Definition $y = \sinh^{-1} x \iff \sinh y = x$; $y = \cosh^{-1} x \iff \cosh y = x$, $y \ge 0$; $y = \tanh^{-1} x \iff \tanh y = x$.

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For every x:
$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$.

PROOF $y = \sinh^{-1} x$: $x = \frac{e^y - e^{-y}}{2}$, so $\left(e^y\right)^2 - 2x\,e^y - 1 = 0$, $e^y = x \pm \sqrt{x^2+1}$. But $e^y > 0$ and $x - \sqrt{x^2+1} < 0$; so $e^y = x + \sqrt{x^2+1}$, and $y = \ln\left(x + \sqrt{x^2+1}\right)$. ∎

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EXAMPLE $\sinh^{-1}\frac{3}{4} = \ln\left(\frac{3}{4} + \sqrt{\frac{9}{16}+1}\right) = \ln\left(\frac{3}{4}+\frac{5}{4}\right) = \ln 2$; check: $\sinh(\ln 2) = \frac{2-\frac{1}{2}}{2} = \frac{3}{4}$.

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EXAMPLE $\sinh^{-1}\frac{3}{4} = \ln\left(\frac{3}{4} + \sqrt{\frac{9}{16}+1}\right) = \ln\left(\frac{3}{4}+\frac{5}{4}\right) = \ln 2$; check: $\sinh(\ln 2) = \frac{2-\frac{1}{2}}{2} = \frac{3}{4}$.

PROOF that $\left(\sinh^{-1} x\right)' = \frac{1}{\sqrt{1+x^2}}$: $\sinh' = \cosh \neq 0$, so $\sinh^{-1}$ is differentiable (the theorem on the derivative of an inverse function). $\sinh y = x$ gives $\cosh y \cdot y' = 1$, and $\cosh y \ge 1$: $y' = \frac{1}{\cosh y} = \frac{1}{\sqrt{1+\sinh^2 y}} = \frac{1}{\sqrt{1+x^2}}$. ∎

A little history · 1638–1757

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1757. **Vincenzo Riccati** introduces the hyperbolic functions, written Sh. and Ch.; soon after, **Johann Heinrich Lambert** develops them systematically and gives them the abbreviations we use. In our notation the curve of 1691 is $y = c + a\cosh\frac{x}{a}$.

Today in three lines

1. $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$: restrict first, then invert — sin to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, cos to $[0, \pi]$, tan to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. The first undoing equation holds only there: $\sin^{-1}(\sin \pi) = 0$.

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3. $\cosh x = \frac{e^x + e^{-x}}{2}$, $\sinh x = \frac{e^x - e^{-x}}{2}$: $\cosh^2 x - \sinh^2 x = 1$; $(\sinh x)' = \cosh x$, $(\cosh x)' = \sinh x$ — no minus sign; $(\tanh x)' = 1 - \tanh^2 x$.

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This week:

Homework: §6.6 and §6.7 — read and work out the examples (the integrals come later). Next: maximum and minimum values (§3.1).

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: "$\frac{d}{dx}\cosh x = \sinh x$, *with no minus sign, because . . . " (one sentence, or one line of algebra).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.